

Article

ESG Ratings, Media Sentiment, and Corporate Innovation Performance

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Abstract: This study employs a dataset comprising Chinese A-share listed companies from 2012 to 2021, with the objective of empirically examining the relationship between corporate ESG ratings, external media sentiment, and corporate innovation performance. The findings indicate that a favourable ESG rating can significantly enhance a company's innovation performance, with external media sentiment exerting a supportive influence. Further analysis indicates that the optimistic media sentiment generated by a favourable ESG rating serves to reduce the financing threshold for companies, alleviate external financing constraints, and ultimately promote innovation performance. Heterogeneity analysis shows that in companies with higher financing constraints and lower levels of marketization in their location, the improvement of ESG ratings has a better effect on enhancing innovation performance. By examining the influence of external media sentiment on the relationship between ESG ratings and a company's innovation performance, this study contributes to the understanding of the mechanisms through which media affects a company's quality development. These findings have implications for the stimulation of a company's innovation vitality and the guidance of their accelerated transformation.

Keywords: ESG rating; financing constraints; innovation performance; media sentiment



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1. Introduction

The decline of the real estate market has brought about an important transition point in China's economic transformation. Concurrently, the prolonged deflationary period and the rising unemployment rate have compelled the most populous developing nation to reassess its development strategy. In order to address these issues, in April 2024, the Chinese government, through the country's three major stock exchanges, officially issued guidance on sustainable development information disclosure. This encouraged listed companies to disclose ESG information in accordance with uniform standards. It is evident that the ESG report disclosure, which is quantifiable and of a uniform calibre, has greater practical significance for the improvement of China's ESG system than social responsibility information, which is perceived as lacking depth and has been subject to criticism. It seems reasonable to posit that the introduction of a unified disclosure standard will intensify the competitive landscape among companies, prompting them to assume greater social responsibilities and pursue continuous improvement through innovation. This is an approach that is entirely consistent with the values of sustainable company development [1].

Nevertheless, it remains to be seen whether the competitive advantage offered by ESG can be effectively transferred to the sustainable development of companies. The mechanism behind this phenomenon requires further investigation. Previous research indicates that a favourable market environment can provide sufficient financial support for a company's business activities through effective external monitoring and evaluation, thereby stimulating company's innovative vitality [2]. It can thus be predicted that in

the context of the Chinese government's recent promotion of ESG concepts, ESG ratings will undoubtedly become a key criterion for external investors to assess the sustainable development of companies [3]. They will also play an important role in guiding the external supervision of the market and the direction of social resources. In light of the aforementioned evidence, it seems reasonable to posit that the monitoring and resource allocation effects of ESG ratings may encourage companies to engage in more sustainable business activities.

In the context of the information economy, the media assumes a significant role as a publisher and interpreter of ESG ratings, functioning as an intermediary in the dissemination of information between the market, stakeholders, and companies. On the one hand, media attention and interpretation can often serve to enhance the monitoring and transmission of information, thereby improving the quality of ESG management of companies and mitigating the problem of information asymmetry among market participants [4]. Conversely, the interpretation of ESG ratings by the media can exert a significant influence on the decision-making of the general public and even investors, thereby affecting the external resource acquisition and future strategic planning of companies [5]. It is evident that the role of the media in the transmission channel of the impact of ESG ratings on corporate development cannot be overlooked. However, in recent years, the majority of research conducted within ESG-related fields has concentrated on the examination of the role of media attention in the context of corporate governance [6], with minimal attention paid to the potential impact of media sentiment. It is evident that in the present circumstances, where the Chinese government is actively promoting the ESG concept, media reports imbued with biased emotions will undoubtedly exert a considerable influence on the general public and investors [7], which may further accentuate the impact of ESG ratings on corporate transformation.

In light of the heightened media scrutiny of company ESG and its associated innovation activities during China's ongoing economic transition, it is crucial to ascertain whether there is a discernible correlation between media sentiment and the two variables. It is regrettable that there is no existing literature investigating this from the perspective of media emotion. In light of the above, this paper takes 2012–2021 A-share listed companies as the research sample with a view to further expanding the research in this field and analysing and discussing the above problems and the mechanism behind them in greater depth.

The theoretical and practical contributions of this study are as follows: (1) In general, media reports that evoke positive or negative emotions frequently challenge the objectivity of their content. Consequently, a considerable number of scholars have concentrated their attention on analysing the adverse economic consequences of media-induced emotions on business companies [8,9]. This article presents a study examining the influence of media sentiments on the propagation of ESG principles in China. It also investigates the potential economic benefits from this phenomenon from the vantage point of corporate innovation. This not only enhances the body of knowledge in related disciplines but also furnishes the government with invaluable insight into the strategic deployment of media resources to advance the quality of corporate development in the future. (2) The relationship between ESG ratings and corporate innovation has been the subject of considerable debate in previous studies, with evidence suggesting that the two may have different impact relationships in different scenarios [10,11]. The present study offers a novel perspective and empirical material to the field, examining the potential impact of ESG ratings on corporate transformation and development in a developing country that is just beginning to vigorously promote ESG concepts and is undergoing economic transformation. Additionally, it investigates the role of media sentiment in this process. This not only provides new empirical evidence on the role of the media in promoting the sustainable development of companies, but also extends research on the external drivers and related pathways of corporate innovation capabilities.

2. Literature Review and Theoretical Hypotheses

2.1. ESG Rating and Corporate Innovation

The extant literature remains inconclusive as to whether ESG ratings exert a positive influence on a company's innovation performance. On the one hand, the vast majority of scholars concur with the theory of competitive strategy that good ESG performance is beneficial for optimising a company's organisational structure and business processes [3,12]. This can provide a favourable competitive and human capital environment for the company's innovation activities. For instance, research has demonstrated that organisations with exemplary ESG performance are more inclined to adopt a low-carbon business model. This not only enhances the quality of products and services but also mitigates the financial implications of environmental infractions that may arise from business operations [13]. This intangible improvement in the competitive environment is a consequence of the aforementioned factors. In a study conducted by Chrichen et al. [1], it was found that organisations with a favourable ESG performance are more likely to demonstrate a commitment to the protection of employees' rights, which in turn helps to attract a greater number of talented individuals. And a robust ESG governance structure can effectively deter managers from engaging in self-serving conduct, thereby promoting greater transparency and reducing agency costs [14]. Furthermore, from the standpoint of corporate reputation, a favourable ESG rating can effectively mitigate a company's reputation risk and accrue greater social trust capital, thereby enabling the company to possess more resources to support its research and innovation activities [10].

Nevertheless, some scholars put forth the proposition of a threshold effect of ESG rating on the enhancement of company value. The argument is that positive returns to the company can only be achieved when ESG rating indicators reach a certain level [15]. For instance, research has demonstrated that a favourable ESG rating does not exert a considerable influence on a company's growth [16]. Conversely, it may even impede the company's financial advancement [17], consequently constraining its R&D investment [10].

The efficacy of ESG ratings in China is contingent upon the incremental standardisation and normalisation of ESG information disclosure by Chinese companies. In particular, third-party organisations are able to conduct multidimensional comprehensive evaluations of a company's environmental, social, and governance performance through the ESG rating system. This system employs publicly available information and materials to categorise a company's ESG performance, assigning ratings that provide insight into the company's sustainability performance. From the perspective of external stakeholders, interest groups such as governments, community members, and investors may utilise ESG rating results to inform the development of intervention plans for future activities of companies [18]. The existing literature provides substantial evidence of a strong correlation between ESG ratings and a company's prospective capacity for sustainable development. This is because the social responsibility undertaken by the company is often contingent upon the external support it receives. This may, to some extent, result in a more effective allocation of market funds to companies that are more inclined to engage in green transformation and technological innovation [19]. Furthermore, companies with superior ESG performance are typically regarded as exhibiting a diminished propensity for environmental infractions, which represents a pivotal consideration for long-term investors [20]. Therefore, it can be reasonably inferred that a favourable ESG rating conveys a positive signal to the external environment, thereby enhancing the standing of the organisation with its stakeholders. Consequently, this may facilitate the acquisition of resources and assistance for the company's research and development and innovation activities [21]. In light of the aforementioned theory, it can be posited that an enhancement in ESG ratings exerts a favourable influence on the innovation performance of companies. Therefore, this study makes the following assumption:

H1. *The improvement of ESG ratings can effectively enhance a company's innovation performance.*

2.2. ESG Ratings, Media Sentiment, and Corporate Innovation

In consequence of the growing salience of ESG concepts in China and the intensified scrutiny from a range of stakeholders, including the media, which serves as an external supervisor and information disseminator of corporate ESG activities, it is inevitable that the ESG performance of companies will be subjected to evaluation and analysis [1]. In related fields of research, a substantial body of literature has examined the positive effects of the media's external supervisory and information dissemination functions on corporate innovation [8]. However, with the rapid growth of the Internet and the exponential increase in the volume of information, scholars have increasingly recognised that, in addition to the efficacy of information dissemination and supervision and governance, media reports imbued with emotive content may also exert a positive or negative influence on the decision-making processes of companies [4]. Such effects may, in turn, impact the research and development (R&D) and innovation activities of companies. In light of the Chinese government's recent advocacy for ESG, it is reasonable to anticipate that shifts in corporate ESG ratings will result in a greater prevalence of reports with emotional impact [2]. It has been demonstrated that the media frequently employs sensationalist techniques to attract a larger audience, a practice that is driven by considerations related to commercial value [7]. In light of the Chinese government's active promotion of ESG, there is a rationale for the media to process and present ESG-related information in a manner aligned with the government's policy orientation [22]. In light of the aforementioned evidence, it seems reasonable to posit that a favourable ESG rating may attract considerable emotional media coverage. In light of the aforementioned evidence, the following hypothesis is therefore proposed:

H2. *The improvement of ESG ratings can effectively drive media sentiment.*

The existence of agency conflicts and financing constraints is frequently identified as a significant impediment to corporate innovation performance in the field of corporate finance [21]. However, existing research indicates that media sentiment may also exert a diametrically opposed influence on a company's innovation performance along these two dimensions [23,24]. On the one hand, some studies indicate that media sentiment can distort the veracity of information, potentially exacerbating external investors' cognitive biases and negatively impacting the normal operations of companies. For instance, research has demonstrated that an excessively optimistic media sentiment exerts a masking effect [25], which intensifies agency conflicts between companies and investors and restrains a company's proclivity to innovate [23]. Conversely, some scholars have indicated that positive media sentiment can facilitate the movement of talent, reduce the financial constraints faced by companies, and support the advancement of R&D activities [26]. To illustrate this, Carlini et al. discovered that optimistic media sentiment can stimulate positive investor sentiment and diminish equity financing costs for companies [24], which is undoubtedly advantageous for research and innovation activities that necessitate substantial and sustained capital investment.

The preceding analysis indicates that, from the perspective of information asymmetry, high media sentiment will serve to exacerbate the cognitive bias of external investors. This, in turn, has the effect of weakening the external supervision and information dissemination functions of the media, thereby creating favourable conditions for opportunistic and self-interested behaviour on the part of management. As a result, the innovative activities of companies are constrained. From the perspective of financing constraints, an optimistic media sentiment may serve to amplify the reputation effect of companies, thereby attracting greater resources and promoting innovation activities. In conclusion, it can be posited that the potential for media sentiment to exert an entirely opposing influence on corporate innovation performance is a possibility that must be considered. In light of this, the following competitive hypotheses are proposed:

H3a. *In the process of ESG ratings driving the innovation performance of companies, media sentiment plays a positive role.*

H3b. *In the process of ESG ratings driving the innovation performance of companies, media sentiment exerts a negative effect.*

3. Methods

3.1. Model Design

In order to verify hypotheses H1–H3, this study employs a bidirectional fixed-effects regression model, as outlined below:

$$Patent_{i,t+1} = \beta_0 + \beta_1 ESGr_{i,t} + \theta' X_{i,t} + \alpha + \gamma + \varepsilon_{i,t+1} \quad (1)$$

$$Tone_{i,t} = \beta_0 + \beta_1 ESGr_{i,t} + \theta' X_{i,t} + \alpha + \gamma + \varepsilon_{i,t} \quad (2)$$

$$Patent_{i,t+1} = \beta_0 + \beta_1 ESGr_{i,t} + \beta_2 ESGr \times Tone_{i,t} + \beta_3 Tone_{i,t} + \theta' X_{i,t} + \alpha + \gamma + \varepsilon_{i,t+1} \quad (3)$$

In the above Equation, i and t , respectively, represent the company and year, in which α represents individual fixed effects used to control for individual omitted features that do not change over time but have not been observed. γ represents time-fixed effects used to control for time trends and macro shocks, and ε represents error terms.

Specifically, Equation (1) is used to examine the $ESGr_{i,t}$ impact of a company's ESG rating on its innovation performance, where the independent variable $ESGr_{i,t}$ is the ESG rating of company i in year t . In light of the potential temporal discrepancy between the influence of ESG ratings on corporate innovation performance and the manifestation of such effects, this study selects the innovation performance $Patent_{i,t+1}$ of company i in year $t + 1$ as the dependent variable. $X_{i,t}$ is a series of control variables that may be related to corporate innovation performance. Equation (2) is used to examine the impact of corporate ESG ratings on external media sentiment, where the dependent variable $Tone_{i,t}$ is the media sentiment of company i in year t , and the independent $ESGr_{i,t}$ and control variables $X_{i,t}$ are consistent with Equation (1). The objective of Equation (3) is to ascertain the influence of media sentiment on the impact of ESG ratings on an organisation's innovation performance. Therefore, on the basis of Equation (1), we add $ESGr \times Tone_{i,t}$ as an interactive variable and control the impact of $Tone_{i,t}$, with the control variable $X_{i,t}$ unchanged.

3.2. Variable Design

3.2.1. Company Innovation Performance

The innovation activities of companies typically progress through a series of stages, including fundraising, research and development investment, patent application, and patent authorisation. This study employs the number of patents held by a company as a proxy for its innovation performance. This is because the patent data of a company are typically more readily accessible, and different patents can be classified into distinct technical fields, facilitating the processing of the data. Secondly, the number of patents held by a company has been identified as a key indicator of technological development in previous studies, offering valuable insights [27]. Accordingly, this study employs the natural logarithm of the number of patents granted (Patent1) and applied for (Patent2) within the year as proxy variables for corporate innovation performance, which can effectively reflect the innovation willingness and ability of companies [2].

3.2.2. ESG Rating

The most commonly used ESG rating indices internationally are the MSCI and Bloomberg ESG rating indices. In China, the most commonly used ESG rating indices are the Huazheng, Bloomberg, and Wind ESG rating indices [28]. In light of the close proximity to the Chinese market, extensive coverage, and high timeliness of Huazheng ESG rating data, this study employs the Huazheng ESG rating index as the explanatory variable. In particular, the Huazheng ESG rating categorises a company's ESG performance

into nine distinct levels, ranging from excellent to poor. The rating scale employed is as follows: AAA, AA, A, BBB, BB, B, CCC, CC, and C. A score of 1 to 9 is assigned to each of the nine levels, from C to AAA. In the robustness test section, this study employs the Bloomberg ESG rating index, which is a widely utilised benchmark both internationally and domestically, as the explanatory variable for robustness testing.

3.2.3. Media Emotion

With regard to the selection of a media sentiment dictionary library, the LM dictionary, developed by Loughran and McDonald [29], is more appropriate for research in the field of corporate finance and is currently the most influential English sentiment dictionary. In China, Jiang et al. [30] have created a Chinese financial text sentiment dictionary based on the LM dictionary and combined it with the word2vec algorithm. The dictionary comprises 3338 positive words, 5890 negative words, and 4448 unique words, rendering it more suitable for the analysis of financial media texts in China.

Secondly, the financial media texts selected in this study encompass both online financial news and newspaper financial news. The corpus comprises online financial news, drawn from 20 leading online financial media outlets, including Hexun, Sina Finance, and Dongfang Fortune. These online financial media outlets are among the most prominent in China in terms of financial news reporting, both in terms of quantity and quality. They are financial websites that Chinese investors frequently consult and follow. The corpus of financial news from newspapers and magazines comprises eight mainstream financial newspapers, including the China Securities Journal, Shanghai Securities News, and the First Financial Daily. These eight major financial newspapers are characterised by the timely, high-quality, and influential reporting of financial news and are also the most commonly used data sources for media research in China's economy, management, and business. Ultimately, in accordance with the methodology proposed by Li Qian et al. [31], we employed text analysis to ascertain the emotional tone slant of each media report and evaluated the media's attention intensity to the enterprise through a three-level scoring system. Ultimately, the media's emotional tone towards the enterprise was constructed based on the aforementioned two indicators. The specific construction method is as follows:

$$Slant_j = \frac{PositiveWords_j - NegativeWords_j}{TotalWords_j} \quad (4)$$

$$Tone_{i,t} = \frac{1}{N} \sum_{j=1}^N Slant_j \times intensity_{i,j} \quad (5)$$

In Equation (4), $Slant_j$ represents the emotional tone of the j -th media report, $PositiveWords_j$ is the number of positive words in the report, $NegativeWords_j$ is the number of negative words in the report, and $TotalWords_j$ is the total number of words in the report. In Equation (5), N represents the number of media reports on company i in year t . $intensity_{i,j}$ indicates the j -th media report's attention to company i . If the name of company i appears in the title of a report, the $intensity_{i,j}$ is assigned a value of 3. If the name of company i appears more than or equal to 3 times in the content of a report, then $intensity_{i,j}$ assigned a value of 2. If the name of company i appears less than 3 times in the content of a report, then $intensity_{i,j}$ is assigned a value of 1. From this, we can obtain the media sentiment of company i in year t . And $Tone_{i,t}$ represents the degree of positivity associated with the media sentiment of company, with a higher value indicating a more favourable assessment.

3.2.4. Control Variables

The control variables $X_{i,t}$ selected for consideration in our study are as follows: company size (Size), debt level (Lev), asset return rate (Roa), listing year (Age), equity concentration (Top1), cash flow level (CFO), and other firm-level control variables related to financial information. Additionally, the variable $Acover$, representing media and analyst

coverage, is employed to control for the impact of external attention on the enterprise. The definitions of the variables are provided in Table 1.

Table 1. Symbols and definitions of variables.

Variable	Symbol	Definition
Enterprise innovation performance	Patent1	The natural logarithm of the number of authorised innovation patent applications
Enterprise innovation performance	Patent2	The natural logarithm of the number of innovative patents
ESG rating	ESGr	ESG rating of the company in the current year
Media sentiment	Tone	The media's sentiment towards reporting on companies in the current year based on Equations (4) and (5)
Enterprise scale	Size	Natural logarithm of total assets
Asset–liability ratio	Lev	Total liabilities/total assets
Return on assets	Roa	Net profit/total assets
Concentration of equity	Top1	Shareholding ratio of the largest shareholder
Cash flow level	CFO	Operating net cash flow/total assets
Age of listing	Age	This is the natural logarithm of the age at which companies become publicly traded
Media attention	Cover	The natural logarithm of the number of media reports
Analyst coverage	Acover	Natural logarithm of the number of analysts tracking companies

3.3. Data Source

The initial sample for this study comprises A-share listed companies from 2012 to 2021. Following the exclusion of companies facing delisting risks and those lacking key financial data, this study obtained panel data comprising 10,200 firm-year observations. The media text materials were obtained from the Infobank database and the CNRDS database, while the financial data of the enterprise were sourced from the CSMAR database. Finally, in order to mitigate the impact of outliers in the data on the research design, a 1% Winsorize treatment was performed on all continuous variables at the firm level.

4. Test Results

4.1. Descriptive Statistics

Table 2 presents the results of the descriptive statistical analysis. A total of 10,200 company observations from 2012 to 2021 were analysed. The mean and median values of Patent1 (Patent2) were found to be 2.813 (3.441) and 2.994 (3.738), respectively, with maximum values of 6.999 and 7.888, respectively, and minimum values of 0. The standard deviations were found to be 1.944 and 2.214, indicating that the sample companies exhibited some differences in innovation performance. The mean and median of the independent variable ESGr are 4.436 and 4.000, respectively, with a maximum value of 8.000, a minimum value of 1.000, and a standard deviation of 1.042. This indicates that there are some differences in ESG ratings among companies, with the majority of companies' ESG ratings being above B. The mean and median of media sentiment tone are 0.097 and 0.101, respectively, with a maximum value of 0.503, a minimum value of −0.419, and a standard deviation of 0.110. This indicates that the majority of media sentiments are optimistic and that there is considerable variation in media sentiment among different companies.

With regard to the control variables, the mean and median of enterprise size (Size) are 23.057 and 22.952, respectively, with maximum and minimum values of 25.936 and 15.938, respectively. This indicates that the samples of companies selected in this study are medium and large companies. The mean and median of media attention (Cover) are 5.807 and 5.741, respectively, with maximum and minimum values of 10.948 and 1.609, respectively, and a standard deviation of 1.118. The mean and median of analyst coverage (Acover) are 2.787 and 2.763, respectively, with maximum and minimum values of 5.404 and 0.549, respectively, and a standard deviation of 0.555. These figures indicate that media attention and analyst coverage vary widely across companies. The remaining control

variables are within a reasonable range of values and are generally consistent with the existing literature [2,21], and thus will not be repeated here.

Table 2. Summary statistics.

Variable	Obs	Mean	Min	Median	Max	SD
Patent1	10,200	2.813	0	2.944	6.999	1.944
Patent2	10,200	3.441	0	3.738	7.888	2.214
ESGr	10,200	4.436	1.000	4.000	8.000	1.042
Tone	10,200	0.097	−0.419	0.101	0.503	0.110
Size	10,200	23.057	19.538	22.952	25.936	1.282
Lev	10,200	0.482	0.057	0.494	0.892	0.198
Age	10,200	2.423	0.693	2.565	3.401	0.588
Roa	10,200	0.053	−0.247	0.043	0.229	0.058
Top1	10,200	0.382	0.090	0.369	0.752	0.161
CFO	10,200	0.111	−0.739	0.092	0.715	0.178
Cover	10,200	5.807	1.792	5.741	10.948	1.118
Acover	10,200	2.787	0.549	2.763	5.404	0.555

4.2. Correlation Analysis

Prior to conducting a regression analysis, it was essential to ascertain the reliability of the research findings presented in this study. This could be achieved by first examining the presence of multicollinearity between the variables under investigation. The Pearson correlation coefficients among the variables were examined in order to achieve this objective, as illustrated in Table 3. The correlation coefficients between ESG rating and variables Patent1 and Patent2 have been found to be statistically significant at the 1% level, which indicates a significant correlation between ESG rating and corporate innovation performance. Similarly, a significant relationship is observed between ESGr and Tone (significant at the 1% level), which indicates a significant correlation between ESG rating and media sentiment. Finally, the correlation coefficients between the control variables selected in this study and Patent are significant at the 10% level, indicating that the control scalar used in this study is reasonable. Furthermore, the Pearson correlation coefficients between the control variables are observed to be less than 0.5, which serves to eliminate, to some extent, the possibility of multicollinearity in the main regression model presented in this study.

Table 3. The correlation matrix.

Variable	Patent1	Patent2	ESGr	Tone	Size	Lev	Age	Roa	Top1	CFO	Cover	Acover
Patent1	1											
Patent2	0.994 ***	1										
ESGr	0.084 ***	0.078 ***	1									
Tone	0.192 ***	0.200 ***	0.060 ***	1								
Size	0.277 ***	0.261 ***	0.172 ***	0.070 **	1							
Lev	0.036 ***	0.021 *	0.013	−0.088 ***	0.489 ***	1						
Age	−0.024 **	−0.031 **	0.022 ***	−0.096 ***	0.257 ***	0.242 ***	1					
Roa	0.033 ***	0.032 ***	0.073 ***	0.192 ***	−0.144 **	−0.456 ***	−0.207 ***	1				
Top1	−0.013 *	−0.018 *	0.023 **	−0.062 ***	0.225 ***	0.086 ***	−0.068 ***	0.055 ***	1			
CFO	−0.077 ***	−0.076 ***	0.001	0.045 ***	0.001	−0.213 ***	0.033 ***	0.284 ***	0.043 ***	1		
Cover	0.174 ***	0.161 ***	0.128 ***	−0.156 ***	0.486 ***	−0.183 ***	0.034 **	0.085 ***	0.061 ***	−0.020 *	1	
Acover	0.176 ***	0.164 ***	0.108 ***	−0.152 ***	0.476 **	−0.170 ***	0.032 ***	0.080 ***	0.052 ***	−0.021 **	0.337 ***	1

Note: ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

4.3. Preliminary Regression Results

Table 4 presents the regression results for Equation (1). As can be observed in columns (1) and (3), the addition of control variables results in a notable increase in the regression coefficients of ESGr for Patent1 and Patent2, reaching 0.034 and 0.038, respectively.

These values are statistically significant at the 1% level, indicating a positive association between the variables in question. The signs and significance of the remaining control variables are largely consistent with the existing literature [2,21]. Among the variables, the coefficient of media attention is significantly positive, which may be due to the fact that an increase in media attention can improve the internal governance level of companies, thereby promoting their innovation willingness [32]. Similarly, an increase in analyst coverage can help to reduce information friction between the company and the outside world, thereby improving the company's innovation performance. Secondly, there is a significant positive correlation between company size and patent count. Larger companies typically have lower financing constraints and therefore have more funds available to support research and innovation activities. The debt ratio (Lev) is significantly inversely proportional to patent, which may be attributed to the fact that companies with higher debt ratios may be more susceptible to financial constraints.

Table 4. Benchmark regression results.

Variable	(1)	(2)	(3)	(4)
	Patent1	Patent1	Patent2	Patent2
ESGr	0.034 *** (3.02)	0.059 *** (3.50)	0.038 *** (2.67)	0.060 *** (4.15)
Size	0.565 *** (12.94)		0.499 *** (13.13)	
Lev	−0.378 ** (−2.05)		−0.271 * (−1.72)	
Age	0.159 (1.62)		0.135 (1.58)	
Roa	0.501 (1.44)		0.471 (1.59)	
Top1	−0.006 ** (−2.33)		−0.005 ** (−2.12)	
CFO	−0.062 (−0.64)		−0.043 (−0.54)	
Cover	0.154 ** (2.11)		0.178 * (1.81)	
Acover	0.332 *** (2.85)		0.367 ** (1.99)	
Constant	−9.719 *** (−9.94)	3.169 *** (41.87)	−8.924 *** (−10.48)	2.538 *** (39.14)
Firm fixed effect	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
N	10,200	10,200	10,200	10,200
Adj-R2	0.828	0.821	0.831	0.826

Note: figures in () are *t* values; ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

In columns (2) and (4), the regression coefficients of ESGr for Patent1 and Patent2 remain significantly positive at the 1% level, at 0.059 and 0.060, respectively, even when control variables are not added. Accordingly, the results presented in Table 4 substantiate the assertion that a commendable ESG rating can markedly enhance a company's innovation performance, thereby corroborating hypothesis H1.

4.4. Endogenous Test

4.4.1. Instrumental Variable Two-Stage Regression (IV-2SLS)

Despite the incorporation of a two-way fixed effect encompassing individual and year composition within Equation (1), which can mitigate the endogenous issue to a degree, there may still be instances where endogenous issues such as missing variables and mutual

causality persist. Consequently, instrumental variables are constructed in order to further alleviate the endogenous problem.

Firstly, our study employs the lag term of ESGR (ESGr_L) and the mean value of ESGR for other companies in the same industry (ESGr_A) as the tool variable, in accordance with the methodology proposed by Ban et al. [2]. This instrumental variable construction method, commonly used in the field of corporate finance, is deemed to satisfy the requisite criteria for exogenous variables. The rationality of the instrumental variables was evaluated, and the findings indicated an insignificant Kleibergen PAAP rk LM statistic ($p = 0.000$), suggesting that the instrumental variables were not affected by unidentifiable issues. Furthermore, the Kleibergen PAAP rk Wald F value ($F = 504.66$) was considerably higher than the critical value at the 10% level of the Stock Yogo test, suggesting a robust correlation between the instrumental variables and the endogenous explanatory variables. The regression results for IV-2SLS are presented in columns (1) to (3) of Table 5.

Table 5. Endogeneity test.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	IV-2SLS		PSM		T + 2		
	ESGr	Patent1	Patent2	Patent1	Patent2	Patent1	Patent2
ESGr		0.138 ** (2.27)	0.157 ** (2.25)	0.031 *** (3.89)	0.040 *** (3.25)	0.043 *** (3.77)	0.048 *** (2.98)
ESGr_L	0.614 *** (50.19)						
ESGr_A	0.861 *** (22.46)						
Size	0.217 *** (16.87)	0.550 *** (24.49)	0.618 *** (24.10)	0.498 *** (9.05)	0.560 *** (8.97)	0.452 *** (12.90)	0.418 *** (11.35)
Lev	−0.871 *** (−11.08)	0.047 (0.39)	0.002 (0.02)	−0.340 (−1.50)	−0.436 * (−1.66)	−0.301 ** (−2.17)	−0.026 (−0.17)
Age	0.001 (0.993)	−0.066 ** (−2.38)	−0.081 ** (−2.56)	0.254* (1.87)	0.283 * (1.82)	0.143 * (1.87)	0.200 ** (2.39)
Roa	0.419 * (1.90)	1.605 *** (5.25)	1.548 *** (4.44)	0.523 (1.06)	0.523 (0.91)	0.475 * (1.84)	0.376 (1.27)
Top1	0.001 * (0.14)	−0.001 (−1.16)	−0.001 (−1.15)	−0.005 (−1.58)	−0.006 (−1.59)	−0.004 ** (−2.05)	−0.002 (−0.80)
CFO	−0.032 (−0.41)	−0.229 ** (−2.48)	−0.256 ** (−2.43)	0.159 (1.43)	0.203 (1.52)	−0.078 (−1.16)	−0.005 (−0.07)
Cover	0.500 *** (8.19)	0.303 *** (3.36)	0.319 *** (3.09)	0.275 ** (1.99)	0.241 (1.51)	0.241 *** (2.76)	0.158 (1.60)
Acover	0.070 *** (7.72)	0.463 ** (2.53)	0.504 ** (2.41)	0.486 * (1.86)	0.422 * (1.73)	0.436 *** (2.66)	0.306 * (1.66)
Constant				−9.364 *** (−7.67)	−10.103 *** (−7.27)	−8.838 *** (−11.22)	−8.040 *** (−9.75)
Chi-sq (1) P	0.000						
F	504.66						
Firm fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	10,200	10,200	10,200	5778	5778	10,200	10,200
Adj-R2	0.326	0.224	0.211	0.859	0.856	0.801	0.810

Note: figures in () are t values; ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

In the initial stage of the regression analysis, as illustrated in column (1) of Table 5, the coefficients associated with the instrumental variables, ESGr_L and ESGr_A, are both

significantly positive at the 1% level. The regression of columns (2) and (3) in the second stage reveals that the coefficients of the independent variable ESGr are 0.138 and 0.157, respectively. These values indicate a significant positive relationship. Furthermore, the regression results of the other control variables are largely consistent with those of Equation (1), indicating that our findings remain robust.

4.4.2. Propensity Score Matching (PSM)

It is possible that there may be passive management of ESG ratings. This could be the result of external policy impacts faced by the industry in which the enterprise is located, or the environmental policy in which the enterprise is located. In order to mitigate the impact of these issues on the findings of this study, we employed the propensity score matching (PSM) method for subsequent regression analysis. In particular, the companies were initially grouped according to region and industry, with those with ESG rating scores equal to or above the median of the sample designated as the experimental group and the remainder as the control group. This ensured that the experimental and control groups were consistent in terms of industry and region, thereby reducing the impact of specific industry and region factors on the regression results of this study. Secondly, in accordance with the methodology proposed by Ban and Zhu [33], we incorporated additional covariates, including market value (MV), asset liability ratio (LEV), return on assets (ROA), revenue growth rate, cash holding ratio (cash), media attention coverage (media), and analyst coverage (Acover), into the PSM process. We employed a 1:1 retrievable nearest neighbour matching with calliper = 0.25 to ensure that the financial fundamentals and external concerns of the experimental and control groups remained as consistent as possible, given the sufficient sample size. The regression results for the processed samples are presented in column (4) and (5) of Table 5. As demonstrated, coefficients of ESGr are 0.031 and 0.040, respectively, and remain significantly positive at the 1% level, which indicates that the results remain robust even after addressing the potential endogenous issues that may arise from differences in regional, industrial, and enterprise fundamentals.

4.4.3. Patent Using T + 2 Period

Considering that a company's R&D and innovation activities usually have a long period [21], the innovation performance in the forward period may better reflect the company's current willingness to innovate; for this reason, this study further examines the effect of ESGr on a company's innovation performance in the T + 2 period. The results are shown in column (6) and (7) of Table 5, and it can be found that the regression coefficients of ESGr are still significantly positive at the 1% level, which again verifies the robustness of the research findings.

4.4.4. Impact of Macro Environment

In light of the severe recession caused by the COVID-19 pandemic, the Federal Reserve and 20 other central banks have proposed unconventional monetary policy measures since March 2020. In response, some central banks have not only actively intervened in their domestic markets but have also strengthened global coordination. Despite the relative weakness of China's monetary policy regulation in comparison to that of Western countries, the implementation of more aggressive interest rate cuts than usual has been observed. To illustrate this, the People's Bank of China initiated a policy interest rate reduction in the fourth quarter of 2019. This entailed a lowering of the reverse repo rate and the medium-term lending facility rate by 5 basis points in November and December 2019, respectively. In the initial six-month period of 2020, the People's Bank of China proceeded to reduce interest rates on two consecutive occasions, with a total reduction of 30 basis points. This resulted in a cumulative decrease of 35 basis points in the MLF rate and the open market operating rate, which contributed to a decline in the overall market interest rate and loan interest rate. Furthermore, with regard to fiscal policy, China's fiscal deficit ratio surpassed 3% for the first time in 2020, reaching 3.6%, and the deficit size increased by CNY 1 trillion

to CNY 3.76 trillion. In light of the above, it is imperative to consider the potential impact of excessive capital flows on corporate innovation investment [34,35].

Moreover, the significant geopolitical changes that occurred during the period under review have resulted in increased uncertainty regarding trade demand in developed markets, creating a challenging environment for China's trade exports [36]. In particular, the rise in trade protectionism across various countries during the course of the COVID-19 pandemic may result in significant restrictions on the export of Chinese products, which could in turn have an adverse impact on the innovation willingness of enterprises.

To verify the reliability of the research results presented in this article, we conducted further regression tests, controlling for external economic policy uncertainty (EPU), environmental uncertainty (Opunad), M2 growth rate (M2), fiscal deficit ratio (BDR), and the trade export growth rate (EX) of enterprises.

In particular, EPU refers to the research of Baker et al. [37], who analysed the *South China Morning Post* and determined the number of articles related to China's economic policy uncertainty by filtering keywords. The index is continuous, time-varying, and comprehensive, thereby enabling the more accurate reflection of short-term and medium-term changes in economic policy uncertainty. Accordingly, it is employed by the majority of scholars to examine the ambiguity of economic policy [38]. In this study, the monthly economic policy uncertainty index is converted by extracting the annual arithmetic mean, and the final economic policy uncertainty index (EPU) is obtained by dividing the obtained annual economic policy uncertainty index by 100 with reference to Pástor and Veronesi [39].

Environmental uncertainty is defined as the degree of change in the external environment of an enterprise, encompassing a multitude of uncertain factors such as macroeconomic trends, industrial structure, and market demand. It is typically quantified by calculating the standard deviation of operating income over the past five years following industry adjustments. In this study, we make use of the methodology proposed by Zollo and Winter [40], utilising sales revenue data from listed companies over the past five years to estimate abnormal sales revenue through the application of Equation (6).

$$Sale_{i,t} = \varphi_0 + \varphi_1 Year_{i,t} + \varepsilon_{i,t} \quad (6)$$

Among them, Sale represents the sales revenue, Year represents the year, and the residual term ε in Equation (6) is the abnormal sales revenue. The standard deviation of the abnormal sales revenue of a listed company in the past 5 years is calculated, and then, it is divided by the average sales revenue of the past 5 years to obtain the unadjusted environmental uncertainty of the listed company. The median unadjusted environmental uncertainty of all listed companies in the same industry in the same year is the industry environmental uncertainty value, and then, the unadjusted environmental uncertainty of the listed company is divided by the industry environmental uncertainty to obtain the industry adjusted environmental uncertainty (Opunad).

Table 6 shows the regression results after controlling for external shocks. There is multicollinearity between the EPU and the time fixed effects, so we do not control for the time dummy variable in columns (1) and (3). In columns (2) and (4), we further control for the time trend term to minimise the potential impact of omitted variables, following the work of Bonaimé et al. [41]. It can be seen that the coefficients of ESGr in each column of Table 6 are significantly positive at the 1% level and are not significantly different from the results of the baseline regression in Table 4. This indicates that our results are still robust after controlling for macroeconomic factors such as economic policy uncertainty, environmental uncertainty, and M2 growth rate.

Table 6. Control of external shocks.

Variable	(1)	(2)	(3)	
	Patent1	Patent1	Patent2	Patent2
ESGr	0.032 *** (2.91)	0.034 *** (2.76)	0.036 *** (2.67)	0.038 *** (2.88)
Size	0.576 *** (13.68)	0.567 *** (13.14)	0.508 *** (13.87)	0.501 *** (13.32)
Lev	−0.464 ** (−2.56)	−0.429 ** (−2.34)	−0.348 ** (−2.24)	−0.317 ** (−2.01)
Age	0.249 *** (2.82)	0.194 ** (1.98)	0.213 *** (2.79)	0.165 * (1.95)
Roa	0.469 (1.35)	0.478 (1.38)	0.442 (1.49)	0.450 (1.52)
Top1	−0.006 ** (−2.26)	−0.006 ** (−2.21)	−0.005 ** (−2.06)	−0.004 ** (−2.01)
CFO	−0.056 (−0.59)	−0.058 (−0.61)	−0.037 (−0.47)	−0.039 (−0.49)
Cover	0.147 (1.31)	0.154 (1.36)	0.171 * (1.76)	0.177 * (1.82)
Acover	0.320 (1.53)	0.329 (1.57)	0.354 ** (1.97)	0.362 ** (2.01)
EPU	0.097 *** (4.31)	0.075 *** (2.75)	0.096 *** (4.92)	0.076 *** (3.23)
Opunad	0.187 *** (4.72)	0.168 *** (3.87)	0.165 *** (4.81)	0.148 *** (3.93)
M2	−0.027 *** (−4.78)	−0.023 *** (−3.52)	−0.023 *** (−4.65)	−0.019 *** (−3.40)
BDR	−0.006 (−0.76)	−0.006 (−0.76)	−0.003 (−0.42)	−0.003 (−0.41)
EX	−0.045 (−1.27)	−0.020 (−0.50)	−0.047 (−1.56)	−0.026 (−0.75)
Time trend		0.049 *** (2.86)		0.056 *** (3.88)
Constant	−11.265 *** (−12.61)	−67.727 *** (−3.51)	−11.323 *** (−13.29)	−88.332 *** (−4.87)
Firm fixed effect	Yes	Yes	Yes	Yes
Year fixed effect	No	No	No	No
N	10,200	10,200	10,200	10,200
Adj-R2	0.781	0.801	0.802	0.813

Note: figures in () are *t* values; ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

4.5. Robustness Test

In practice, it is challenging to ascertain the time and conversion rate for the innovation activities of companies that are to be converted into patents. Accordingly, this study employs the R&D investment (RD) of companies over a specified period as a proxy variable for their innovation performance for the robustness test, and RD is defined as the natural logarithm of R&D investment and amount. Secondly, as previously stated, this study employs the internationally and domestically recognised Bloomberg ESG rating index (ESGr_B) as the explanatory variable in the robustness test. The regression results are presented in Table 7.

Table 7. Robustness tests.

Variable	(1)	(2)	(3)
	RD	Patent1	Patent2
ESGr	0.011 *** (8.53)		
ESGr_B		0.006 *** (2.84)	0.007 ** (2.49)
Size	0.000 (0.32)	0.571 *** (13.14)	0.506 *** (13.37)
Lev	−0.027 ** (−2.20)	−0.394 ** (−2.15)	−0.288 * (−1.84)
Age	−0.005 (−1.60)	0.166 * (1.68)	0.142 * (1.66)
Roa	−0.056 *** (−4.39)	0.479 (1.38)	0.445 (1.50)
Top1	−0.000 (−0.91)	−0.006 ** (−2.39)	−0.005 ** (−2.21)
CFO	−0.004 (−0.37)	−0.066 (−0.68)	−0.047 (−0.59)
Cover	0.004 * (1.66)	0.158 (1.38)	0.182 * (1.85)
Acover	0.003 (0.64)	0.344 ** (2.11)	0.380 ** (2.06)
Constant	0.038 (1.18)	−9.861 *** (−10.01)	−9.092 *** (−10.60)
Firm fixed effect	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes
N	10,200	10,200	10,200
Adj-R2	0.785	0.828	0.834

Note: figures in () are *t* values; ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

In column (1) of Table 7, the regression coefficient of ESGr is 0.011, which is significantly positive at the 1% level, indicating that an improvement in ESG rating can effectively increase R&D investment by companies. In columns (2) and (3), when ESGr_B is taken as the independent variable, the regression coefficients corresponding to Patent1 and Patent2 are 0.006 and 0.007, respectively, and are significant at least at the 5% level, thereby verifying the robustness of the results of this study.

5. Mechanism Test

5.1. The Role of Media Emotions

5.1.1. ESG Rating and Media Sentiment

The preliminary confirmation of hypothesis H1, namely that ESG rating can significantly enhance a company's innovation performance, is achieved through regression analysis of Equation (1). In the theoretical analysis section, it is proposed that improvements in ESG ratings may serve to enhance media sentiment and facilitate the dissemination of more positive reports concerning companies. In order to verify hypothesis H2, we will test Equation (2). The regression results are presented in Table 8. Column (1) reveals that the regression coefficient for ESG is 0.008, a figure that is significantly positive at the 1% level. This finding corroborates the hypothesis that an improvement in ESG rating exerts a substantial influence on media sentiment, thereby substantiating hypothesis H2.

Table 8. The role of media emotions.

Variable	(1)	(2)	(3)
	Tone	Patent1	Patent2
ESGr	0.008 *** (5.61)	0.037 ** (2.20)	0.038 *** (3.63)
ESGr × Tone		0.015 *** (2.78)	0.018 *** (3.01)
Tone		0.205 * (1.67)	0.195 * (1.69)
Size	0.002 (0.46)	0.499 *** (13.14)	0.565 *** (12.94)
Lev	0.045 *** (3.18)	−0.279 * (−1.77)	−0.387 ** (−2.10)
Age	−0.025 *** (−3.29)	0.140 (1.64)	0.164 * (1.67)
Roa	0.608 *** (20.26)	0.359 (1.16)	0.386 (1.07)
Top1	−0.000 (−0.71)	−0.005 ** (−2.11)	−0.006 ** (−2.32)
CFO	0.003 (0.38)	−0.043 (−0.54)	−0.062 (−0.65)
Cover	−0.003 (−0.35)	0.179 * (1.82)	0.155 (1.36)
Acover	−0.021 (−1.14)	0.363 ** (1.97)	0.328 * (1.74)
Constant	0.118 (1.54)	−8.947 *** (−10.52)	−9.742 *** (−9.97)
Firm fixed effect	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes
N	10,200	10,200	10,200
Adj-R2	0.510	0.834	0.828

Note: figures in () are *t* values; ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

5.1.2. The Role of Media Sentiment

In order to verify hypothesis H3, which examines the role of media sentiment in the relationship between ESG ratings and corporate innovation performance, a regression analysis was conducted on Equation (3). The regression outcomes are presented, respectively, in columns (2) and (3) of Table 8. The regression coefficients of the interaction term ESGr × Tone have been observed to be 0.015 and 0.018, respectively. Both coefficients are significantly positive at the 1% level. Similarly, the regression coefficients for ESGr and Tone in the case of Patent are significantly positive at least at the 10% level. In consideration of the aforementioned findings, it can be concluded that media sentiment exerts an influence on the relationship between ESG ratings and corporate innovation performance, thereby substantiating hypothesis H3a.

5.2. Media Sentiment and Corporate Financing Constraints

The regressions presented in Equations (2) and (3) provide evidence supporting hypotheses H2 and H3a, which posit that media sentiment can positively influence the innovative performance of companies driven by ESG ratings. In the theoretical analysis section, we put forth the argument that this positive role may be attributed to the mitigating effect of enhanced media sentiment on a company's financing constraints. Consequently, this study proceeds to investigate the relationship between media sentiment and a company's financing constraints in order to test this conjecture. In particular, this study examines the impact of media sentiment on three key financial indicators: a company's interest on

loans (Cost), the amount of credit obtained from banks (Banklev), and the KZ index. These indicators are examined in the context of the findings presented by Ban and Zhu [32]. The ratio of interest expense to the average of long- and short-term liabilities (Cost) and the ratio of bank borrowing to the company's total assets (Banklev) are both reliable indicators of a company's financing ability. The KZ index, in contrast, is calculated by Kaplan and Zingales [42] from a company's internal financial data, including operating cash flow, cash holdings, cash-out levels, debt levels, and Tobin's Q. It is widely acknowledged as an effective indicator of a company's capacity to finance its operations. Financial data, currently regarded as a more reliable indicator of corporate financing constraints, are used to compute the KZ index. In accordance with the aforementioned information, the following models were constructed for the purpose of testing:

$$Cost_{i,t} = \beta_0 + \beta_1 Tone_{i,t} + \theta' X_{i,t} + \alpha + \gamma + \varepsilon_{i,t} \quad (7)$$

$$Banklev_{i,t} = \beta_0 + \beta_1 Tone_{i,t} + \theta' X_{i,t} + \alpha + \gamma + \varepsilon_{i,t} \quad (8)$$

$$KZ_{i,t} = \beta_0 + \beta_1 Tone_{i,t} + \theta' X_{i,t} + \alpha + \gamma + \varepsilon_{i,t} \quad (9)$$

The regression results for Equations (7)–(9) are presented below. The coefficients of Tone in columns (1), (2), and (3) of Table 9 are significantly negative at the 1% level, significantly positive at the 5% level, and significantly negative at the 5% level, respectively. This suggests that positive media sentiment can assist companies in obtaining greater returns with lower interest rates on bank loans (Banklev) and in reducing their external financing constraints. In light of the aforementioned findings, it can be posited that an improvement in media sentiment can effectively enhance a company's financing ability, which in turn is conducive to enhancing its innovation performance.

Table 9. Financing constraints.

Variable	(1)	(2)	(3)
	Cost	Banklev	KZ
Tone	−0.014 *** (−5.51)	0.012 ** (2.11)	−0.162 ** (−2.39)
Size	−0.019 *** (−5.31)	0.002 * (1.69)	−0.225 *** (−15.69)
Lev	0.008 *** (3.14)	0.465 *** (80.57)	5.332 *** (81.35)
Age	0.005 *** (6.30)	−0.016 *** (−12.68)	0.276 *** (11.22)
Roa	−0.002 (−0.21)	−0.234 *** (−14.90)	−8.755 *** (−42.6)
Top1	−0.001 *** (−3.75)	−0.001 *** (−7.10)	−0.004 *** (−9.79)
CFO	−0.000 (−0.02)	−0.022 *** (−4.38)	−3.423 *** (−16.24)
Cover	0.000 (0.21)	−0.007 *** (−2.68)	−0.008 (−0.34)
Acover	0.011 (0.80)	0.016 *** (3.95)	0.052 (1.29)
Constant	0.003 (1.16)	0.011 (1.64)	0.205 *** (3.07)
Firm fixed effect	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes
N	10,200	10,200	10,200
Adj-R2	0.0421	0.582	0.575

Note: figures in () are *t* values; ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

5.3. Heterogeneity Analysis

5.3.1. Nature of Ownership of the Enterprise

In China, non-state-owned companies (NSOEs) typically encounter greater difficulties in securing loans from banks than state-owned companies (SOEs). This is due to the fact that China's banking system is predominantly state-controlled, which inevitably results in a greater tendency to favour state-owned companies when it comes to loan credit [33]. It is thus anticipated that the intensification of financing constraints resulting from the ESGr upgrade will exert a more pronounced marginal influence on the innovation performance of non-state-owned companies. In order to test this hypothesis, the sample was divided into two groups: state-owned companies and non-state-owned companies. Subsequently, a regression analysis was conducted for each group separately. The regression results are presented in Table 10. The regression coefficients for ESGr in both the Soe and NSoe sample groups are statistically significant at the 1% level for Patent 1 and Patent 2. This indicates that an enhancement in ESGr may result in a discernible improvement in the innovation performance of both state-owned and non-state-owned companies.

Table 10. Heterogeneity of nature of ownership of the enterprise.

Variable	(1)	(2)	(3)	(4)
	Soe	NSoe	Soe	NSoe
	Patent1	Patent1	Patent2	Patent2
ESGr	0.115 *** (5.33)	0.162 *** (7.34)	0.131 *** (5.30)	0.203 *** (6.96)
Size	0.601 *** (24.67)	0.417 *** (12.94)	0.678 *** (24.07)	0.461 *** (12.72)
Lev	−0.097 (−0.65)	0.334 * (1.84)	−0.121 (−0.70)	0.302 (1.46)
Age	−0.071 (−1.59)	−0.030 (−0.71)	−0.085 * (−1.65)	−0.037 (−0.76)
Roa	1.583 *** (3.14)	1.706 *** (4.01)	1.686 *** (2.88)	1.614 *** (3.37)
Top1	−0.001 (−0.73)	−0.004 *** (−2.64)	−0.001 (−0.69)	−0.005 *** (−2.85)
CFO	−0.232 * (−1.81)	−0.079 (−0.54)	−0.288 * (−1.89)	−0.041 (−0.24)
Cover	0.160 (1.60)	0.411 *** (2.94)	0.160 (1.36)	0.452 *** (2.94)
Acover	0.218 (1.07)	−0.609 ** (−2.13)	0.242 (1.00)	−0.676 ** (−2.16)
Constant	−11.921 *** (−25.56)	−7.971 *** (−12.94)	−13.063 *** (−24.19)	−8.342 *** (−12.03)
Firm fixed effect	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
N	3954	6246	3954	6246
Adj-R2	0.623	0.644	0.618	0.628
p-value	0.001 ***		0.002 ***	

Note: figures in () are *t* values; ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. “*p*-values” were used to test for differences in CSR coefficients between groups and were obtained by bootstrap sampling 1000 times.

Further differential tests on the coefficients of grouped samples were performed using the bootstrap method, which yielded empirical *p*-values of 0.001 for columns (1) and (2) and 0.002 for columns (3) and (4). Both of these values were found to be significant at the 1% level. This result indicates a statistically significant difference in grouping coefficients, with column (1) exhibiting a significantly lower ESGr regression coefficient than column

(2) (0.115, significantly less than 0.162), and column (3) displaying a significantly lower ESGr regression coefficient than column (4) (0.131, significantly less than 0.203). Therefore, the results in Table 10 demonstrate the stronger marginal enhancement effect of rising ESGr on the innovation performance of non-state-owned companies relative to state-owned companies, which also provides secondary evidence that the ameliorating effect of financing constraints is an important mechanism for improving a company's innovation performance.

5.3.2. Regional Marketization Process

The results of previous empirical research indicate that the level of marketisation in a region is positively correlated with the quality of the financing environment for companies. This is due to the fact that an improvement in marketisation can serve to reduce the information asymmetry that exists between companies and banks, thereby facilitating the process of obtaining credit for companies [43]. It is therefore anticipated that the relaxation of financial constraints will exert a more pronounced marginal ameliorative effect on corporate innovation in regions characterised by lower levels of marketisation. In order to verify this hypothesis, this study divides the sample into two sub-samples, designated as 'high marketisation' (HMS) and 'low marketisation' (LMS), based on the median score of the marketisation process (MS) of the location where the enterprise is located. Subsequently, a regression analysis is performed on each sub-sample. The Fan Gang marketisation index of each province is used to measure MS, with a higher MS value indicating a higher regional marketisation index. This index is a commonly employed measurement in the field of contemporary Chinese economics.

The regression results are presented in Table 11. The regression coefficients for ESG are found to be significantly positive in both the HMS and LMS samples, indicating that an improvement in ESG ratings can effectively enhance the innovation performance of companies in both regions with high and low marketisation processes. Further differential tests on the coefficients of grouped samples were performed using the bootstrap method, which yielded empirical *p*-values of 0.032 for columns (1) and (2) and 0.045 for columns (3) and (4). Both of these values were found to be significant at the 5% level. This result indicates a statistically significant difference in grouping coefficients, with column (1) exhibiting a significantly lower ESGr regression coefficient than column (2) (0.109, significantly less than 0.186), and column (3) displaying a significantly lower ESGr regression coefficient than column (4) (0.124, significantly less than 0.193). Therefore, the results in Table 10 indicate that in regions with lower marketisation processes, the increase in ESGr has a stronger marginal effect on the innovation performance of companies. This verifies that the improvement effect of financing constraints is an important mechanism for promoting a company's innovation performance once again.

Table 11. Heterogeneity of the regional marketization process.

Variable	(1)	(2)	(3)	(4)
	HMS	LMS	HMS	LMS
	Patent1	Patent1	Patent2	Patent2
ESGr	0.109 *** (5.01)	0.186 *** (7.72)	0.124 *** (4.96)	0.193 *** (7.46)
Size	0.631 *** (23.49)	0.465 *** (16.13)	0.703 *** (22.70)	0.535 *** (16.41)
Lev	−0.101 (−0.59)	0.286 * (1.67)	−0.098 (−0.49)	0.167 (0.85)
Age	−0.053 (−1.24)	−0.089 ** (−2.03)	−0.055 (−1.11)	−0.124 ** (−2.51)
Roa	1.729 *** (3.70)	1.579 *** (3.14)	1.678 *** (3.12)	1.591 *** (2.79)

Table 11. Cont.

Variable	(1)	(2)	(3)	(4)
	HMS	LMS	HMS	LMS
	Patent1	Patent1	Patent2	Patent2
Top1	−0.003 * (−1.80)	−0.001 (−0.86)	−0.004 ** (−2.10)	−0.001 (−0.62)
CFO	−0.178 (−1.33)	−0.478 *** (−3.17)	−0.202 (−1.27)	−0.560 *** (−3.14)
Cover	0.566 *** (4.93)	0.060 (0.53)	0.616 *** (4.69)	0.094 (0.72)
Acover	1.115 *** (4.70)	0.072 (0.32)	1.250 *** (4.60)	0.003 (0.01)
Constant	−12.229 *** (−24.64)	−9.065 *** (−16.69)	−13.194 *** (−23.02)	−9.984 *** (−16.24)
Firm fixed effect	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
N	5088	5088	5088	5088
Adj-R2	0.595	0.579	0.590	0.599
<i>p</i> -value	0.032 **		0.045 **	

Note: figures in () are *t* values; ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. “*p*-values” were used to test for differences in CSR coefficients between groups and were obtained by bootstrap sampling 1000 times.

6. Discussion

In recent years, the Chinese government has demonstrated a notable enthusiasm for the implementation of the ESG concept, which has also led to a significant expansion in the scope of scholarly research on the localisation of ESG in China. It is evident that the media plays a pivotal role in ESG research pertaining to the high-quality development of companies. However, the current research predominantly focuses on the supervisory impact of the media as an external regulator of companies or the intermediary function of the media in the dissemination of information [44,45].

This study differs from the existing literature in that it examines the role of media sentiment in the ESG field from the perspective of media sentiment. It finds that media sentiment plays an important role in improving the external financing constraints of companies and enhancing the innovation willingness of companies. Consequently, the assumptions H1 and H2A are also corroborated. Evidently, our research findings contradict the assertion that the information distortion effect of media sentiment is detrimental to company development. This provides further evidence of the positive economic consequences of media sentiment, which may offer a novel perspective for related research.

Furthermore, it should be noted that this paper is not without limitations. Firstly, it should be noted that China’s economic system differs from that of developed countries. Consequently, the current ESG rating method may not be suitable for evaluating Chinese companies. Consequently, this paper is still constrained by limitations in the construction of ESG evaluation indicators, which may not fully reflect the performance of companies in terms of ESG. Secondly, media sentiment may also exert a complex influence on the development of listed companies via alternative pathways, including the secondary market. Due to the limitations of space, these impact paths have not been investigated in this paper. In the future, it may be possible to consider how to construct a more reasonable ESG evaluation method, including the assignment of a more reasonable scoring weight for the environment, society, and governance to companies in different industries. With regard to the impact path of company development, factors such as the impact of macroeconomic

fluctuations and environmental policy changes need to be further included in the analysis and research.

7. Conclusions and Implications

The findings of this study demonstrate that an enhanced ESG rating can effectively enhance a company's innovative capacity, with media sentiment exerting a significant influence in this regard. This is due to the fact that favourable media sentiment resulting from an exemplary ESG rating can assist companies in securing greater access to bank loans at a reduced interest rate, thereby markedly enhancing their financial capabilities. The aforementioned research conclusions offer the following insights: (1) within the domain of ESG research, media sentiment represents a crucial avenue for enhancing company innovation, underscoring the beneficial impact of media sentiment on company development. It would therefore seem prudent to treat the role of media emotion dialectically, with a view to providing the government with a new tool for guiding the development of companies in the future. (2) In China, small- and medium-sized private companies continue to encounter significant challenges in accessing affordable and accessible financing. If we can effectively utilise the influence of media sentiment, it will assist in motivating these companies to engage in ESG activities and stimulate their willingness to innovate. (3) It is recommended that the government adopt a reasonable regulatory approach to guide the media in their reporting, ensuring that the facts are not exaggerated or distorted, and that a transparent and efficient market information environment is created.

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